INFS3701 Twelve-Month Cost Estimate

Corporate Cloud Storage Gateway · **us-east-1 (N. Virginia)** · On-Demand

Estimate for the optimised architecture (single NAT, small instance classes, low demo traffic). Unit prices from AWS public pricing / **AWS Pricing Calculator** (https://calculator.aws). Ex-tax planning figures.

Assumptions (keep these if you rebuild in calculator.aws)

- 730 hours/month × 12 months; region us-east-1; On-Demand only.
- 1 NAT Gateway (not 2); ~10 GB/month NAT data processing.
- 1 ALB; ~1 LCU average (academic/demo traffic).
- 2 × t3.micro Linux app servers; 8 GiB gp3 each; no detailed monitoring.
- RDS PostgreSQL Multi-AZ db.t3.micro; 20 GiB gp3; no Performance Insights.
- Light S3/Secrets/CloudWatch/SNS bundled as one small ops line (not the cost drivers).
- No Route 53 domain, no Reserved Instances/Savings Plans.

Monthly breakdown (important cost drivers)

Component	Basis (AWS list price)	USD/mo
NAT Gateway (1×, optimised)	\$0.045/hr × 730 + \$0.045 × 10 GB data	33.30
Public IPv4 (NAT + ALB)	3 × \$0.005/hr × 730	10.95
Application Load Balancer	\$0.0225/hr × 730 + ~1 LCU × \$0.008/hr	22.27
EC2 t3.micro × 2 (app tier)	2 × \$0.0104/hr × 730	15.18
EBS gp3 8 GiB × 2	16 GiB × \$0.08/GB-month	1.28
RDS PostgreSQL db.t3.micro Multi-AZ	~\$0.034/hr × 730	24.82
RDS gp3 storage 20 GiB Multi-AZ	20 × \$0.23/GB-month	4.60
Secrets Manager + CloudWatch + SNS (light)	3 secrets + 1 dashboard + light logs/alarms	4.70
	Total per month	**117.10**
	Total for 12 months	**1,405.19**

Optimisation vs a dual-NAT design

This estimate uses **one NAT Gateway** so both private app AZs egress via a single AZ. That cuts roughly **$37/month (~$443/year)** versus two NATs, while keeping two app AZs, Multi-AZ RDS, and the ALB. Disclose the egress trade-off in the design document. Further control for Learner Lab: deploy → capture evidence → **delete the stack** (do not leave NAT/ALB/RDS running).

Outcome for the appendix

≈ **$117 USD per month** · ≈ **$1,405 USD for 12 months** (us-east-1, ex-tax). If the tutor requires a calculator share link, enter only the rows above at calculator.aws and export/share — do not add unused services.

Sources: AWS Pricing Calculator; VPC/NAT, ELB, EC2, RDS, S3, Secrets Manager, CloudWatch public pricing (us-east-1). Estimate only — actual fees depend on usage.